

Assognment_5

ALL files for this assignment can be found at https://github.com/Dylan-Hamilton3412/64060_-Dhamil28

Install and library required packages for this assignment

```
library(ISLR)
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

```
library(cluster)
```

```
## Warning: package 'cluster' was built under R version 4.5.3
```

```
library(stats)
```

```
library(readr)
```

```
library(class)
```

```
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.2
```

```
## Loading required package: ggplot2
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
## Loading required package: lattice
```

Set a seed to ensure reproducibility

```
set.seed(123)
```

Import the cereal data set and examine the variables

```
Cereal_Data <- read.csv("Cereals.csv")
```

```
summary(Cereal_Data)
```

```
##      name                mfr                type                calories
## Length:77              Length:77          Length:77          Min.   : 50.0
## Class :character       Class :character   Class :character       1st Qu.:100.0
## Mode  :character       Mode  :character   Mode  :character       Median :110.0
##                                     Mean   :106.9
##                                     3rd Qu.:110.0
##                                     Max.   :160.0
##
```

```
##      protein      fat      sodium      fiber
## Min.   :1.000   Min.   :0.000   Min.    :  0.0   Min.    : 0.000
## 1st Qu.:2.000   1st Qu.:0.000   1st Qu.:130.0   1st Qu.: 1.000
## Median :3.000   Median :1.000   Median :180.0   Median : 2.000
## Mean   :2.545   Mean    :1.013   Mean    :159.7   Mean    : 2.152
## 3rd Qu.:3.000   3rd Qu.:2.000   3rd Qu.:210.0   3rd Qu.: 3.000
## Max.   :6.000   Max.    :5.000   Max.    :320.0   Max.    :14.000
##
##      carbo      sugars      potass      vitamins
## Min.    : 5.0    Min.    : 0.000   Min.    : 15.00   Min.    :  0.00
## 1st Qu.:12.0    1st Qu.: 3.000   1st Qu.: 42.50   1st Qu.: 25.00
## Median :14.5    Median : 7.000   Median : 90.00   Median : 25.00
## Mean    :14.8    Mean     : 7.026   Mean     : 98.67   Mean     : 28.25
## 3rd Qu.:17.0    3rd Qu.:11.000   3rd Qu.:120.00   3rd Qu.: 25.00
## Max.    :23.0    Max.     :15.000   Max.     :330.00   Max.     :100.00
## NA's    :1      NA's     :1      NA's     :2
##      shelf      weight      cups      rating
## Min.    :1.000   Min.    :0.50    Min.    :0.250   Min.    :18.04
## 1st Qu.:1.000   1st Qu.:1.00    1st Qu.:0.670   1st Qu.:33.17
## Median :2.000   Median :1.00    Median :0.750   Median :40.40
## Mean    :2.208   Mean     :1.03    Mean     :0.821   Mean     :42.67
## 3rd Qu.:3.000   3rd Qu.:1.00    3rd Qu.:1.000   3rd Qu.:50.83
## Max.    :3.000   Max.     :1.50    Max.     :1.500   Max.     :93.70
##
```

From summarizing the data we can see there are a few NA variables located in the Data set

Remove cereals with missing values to ensure all observations can be clustered

```
Cereal_Data <- na.omit(Cereal_Data)
summary(Cereal_Data)
```

```
##      name      mfr      type      calories
## Length:74     Length:74     Length:74     Min.    : 50
## Class :character Class :character Class :character 1st Qu.:100
## Mode  :character Mode  :character Mode  :character Median :110
##                                     Mean    :107
##                                     3rd Qu.:110
##                                     Max.    :160
##      protein      fat      sodium      fiber      carbo
## Min.    :1.000   Min.    :0    Min.    :  0.0   Min.    : 0.000   Min.    : 5.0
## 1st Qu.:2.000   1st Qu.:0    1st Qu.:135.0   1st Qu.: 0.250   1st Qu.:12.0
## Median :2.500   Median :1    Median :180.0   Median : 2.000   Median :14.5
## Mean    :2.514   Mean     :1    Mean    :162.4   Mean     : 2.176   Mean     :14.7
##
```

```
## 3rd Qu.:3.000 3rd Qu.:1 3rd Qu.:217.5 3rd Qu.: 3.000 3rd Qu.:17.0
0
## Max. :6.000 Max. :5 Max. :320.0 Max. :14.000 Max. :23.0
0
## sugars potass vitamins shelf
## Min. : 0.000 Min. : 15.00 Min. : 0.00 Min. :1.000
## 1st Qu.: 3.000 1st Qu.: 41.25 1st Qu.: 25.00 1st Qu.:1.250
## Median : 7.000 Median : 90.00 Median : 25.00 Median :2.000
## Mean : 7.108 Mean : 98.51 Mean : 29.05 Mean :2.216
## 3rd Qu.:11.000 3rd Qu.:120.00 3rd Qu.: 25.00 3rd Qu.:3.000
## Max. :15.000 Max. :330.00 Max. :100.00 Max. :3.000
## weight cups rating
## Min. :0.500 Min. :0.2500 Min. :18.04
## 1st Qu.:1.000 1st Qu.:0.6700 1st Qu.:32.45
## Median :1.000 Median :0.7500 Median :40.25
## Mean :1.031 Mean :0.8216 Mean :42.37
## 3rd Qu.:1.000 3rd Qu.:1.0000 3rd Qu.:50.52
## Max. :1.500 Max. :1.5000 Max. :93.70
```

Convert categorical variables mfr and type into dummy variables

```
Dummy_vars <- dummyVars(~ mfr+type, data = Cereal_Data)
Cereal_Dummy_Var <- predict(Dummy_vars, newdata = Cereal_Data)
## Add the dummy variables back into the original data set
Cereal_Data <- cbind(Cereal_Data, Cereal_Dummy_Var)
```

Remove the original categorical variables columns

```
Cereal_Data$mfr = NULL
Cereal_Data$type = NULL
```

Use the cereal names as row labels so that the names are preserved

```
row.names(Cereal_Data) <- Cereal_Data[,1]
Cereal_Data <- Cereal_Data[,-1]
```

Normalize the data using Z-score in order to not have one variable dominate the clustering algorithm

```
Cereal_Data <- scale(Cereal_Data)
Cereal_Data <- as.data.frame(Cereal_Data)
summary(Cereal_Data)
```

```
## calories protein fat sodium
## Min. :-2.8738 Min. :-1.40687 Min. :-0.9932 Min. :-1.9616
## 1st Qu.: -0.3541 1st Qu.: -0.47733 1st Qu.: -0.9932 1st Qu.: -0.3306
## Median : 0.1498 Median : -0.01256 Median : 0.0000 Median : 0.2131
## Mean : 0.0000 Mean : 0.00000 Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: 0.1498 3rd Qu.: 0.45221 3rd Qu.: 0.0000 3rd Qu.: 0.6661
## Max. : 2.6695 Max. : 3.24083 Max. : 3.9729 Max. : 1.9045
## fiber carbo sugars potass
## Min. :-0.89778 Min. :-2.50014 Min. :-1.6306 Min. :-1.1783
## 1st Qu.: -0.79462 1st Qu.: -0.70143 1st Qu.: -0.9424 1st Qu.: -0.8079
## Median : -0.07249 Median : -0.05903 Median : -0.0248 Median : -0.1201
```

```
## Mean : 0.00000 Mean : 0.00000 Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: 0.34015 3rd Qu.: 0.58337 3rd Qu.: 0.8928 3rd Qu.: 0.3031
## Max. : 4.87925 Max. : 2.12512 Max. : 1.8104 Max. : 3.2660
## vitamins shelf weight cups
## Min. :-1.3032 Min. :-1.4617 Min. :-3.4600 Min. :-2.4251
## 1st Qu.: -0.1818 1st Qu.: -1.1612 1st Qu.: -0.2008 1st Qu.: -0.6432
## Median : -0.1818 Median : -0.2599 Median : -0.2008 Median : -0.3038
## Mean : 0.0000 Mean : 0.0000 Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: -0.1818 3rd Qu.: 0.9420 3rd Qu.: -0.2008 3rd Qu.: 0.7568
## Max. : 3.1822 Max. : 0.9420 Max. : 3.0583 Max. : 2.8780
## rating mfrA mfrG mfrK
## Min. :-1.7336 Min. :-0.1162 Min. :-0.646 Min. :-0.667
## 1st Qu.: -0.7071 1st Qu.: -0.1162 1st Qu.: -0.646 1st Qu.: -0.667
## Median : -0.1510 Median : -0.1162 Median : -0.646 Median : -0.667
## Mean : 0.0000 Mean : 0.0000 Mean : 0.000 Mean : 0.000
## 3rd Qu.: 0.5807 3rd Qu.: -0.1162 3rd Qu.: 1.527 3rd Qu.: 1.479
## Max. : 3.6578 Max. : 8.4861 Max. : 1.527 Max. : 1.479
## mfrN mfrP mfrQ mfrR
## Min. :-0.2674 Min. :-0.3696 Min. :-0.321 Min. :-0.321
## 1st Qu.: -0.2674 1st Qu.: -0.3696 1st Qu.: -0.321 1st Qu.: -0.321
## Median : -0.2674 Median : -0.3696 Median : -0.321 Median : -0.321
## Mean : 0.0000 Mean : 0.0000 Mean : 0.000 Mean : 0.000
## 3rd Qu.: -0.2674 3rd Qu.: -0.3696 3rd Qu.: -0.321 3rd Qu.: -0.321
## Max. : 3.6896 Max. : 2.6692 Max. : 3.073 Max. : 3.073
## typeC typeH
## Min. :-8.4861 Min. :-0.1162
## 1st Qu.: 0.1162 1st Qu.: -0.1162
## Median : 0.1162 Median : -0.1162
## Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: 0.1162 3rd Qu.: -0.1162
## Max. : 0.1162 Max. : 8.4861
```

Compute the Euclidean distance matrix between cereals

```
Data_dist <- dist(Cereal_Data, method = "euclidean")
```

Part A

Compare four linkage methods using AGNES hierarchical clustering

##Single Linkage

```
HClust_Single <- agnes(Cereal_Data, method = "single")
```

```
HClust_Single$ac
```

```
## [1] 0.8543243
```

Complete Linkage

```
HClust_Complete <- agnes(Cereal_Data, method = "complete")
```

```
HClust_Complete$ac
```

```
## [1] 0.8618592
```

```
## Average Linkage
HClust_Average <- agnes(Cereal_Data, method = "average")
HClust_Average$ac
```

```
HClust_Average <- agnes(Cereal_Data, method = "average")
HClust_Average$ac
```

```
## [1] 0.8518632
```

```
## Ward's Linkage
HClust_Ward <- agnes(Cereal_Data, method = "ward")
HClust_Ward$ac
```

```
HClust_Ward <- agnes(Cereal_Data, method = "ward")
HClust_Ward$ac
```

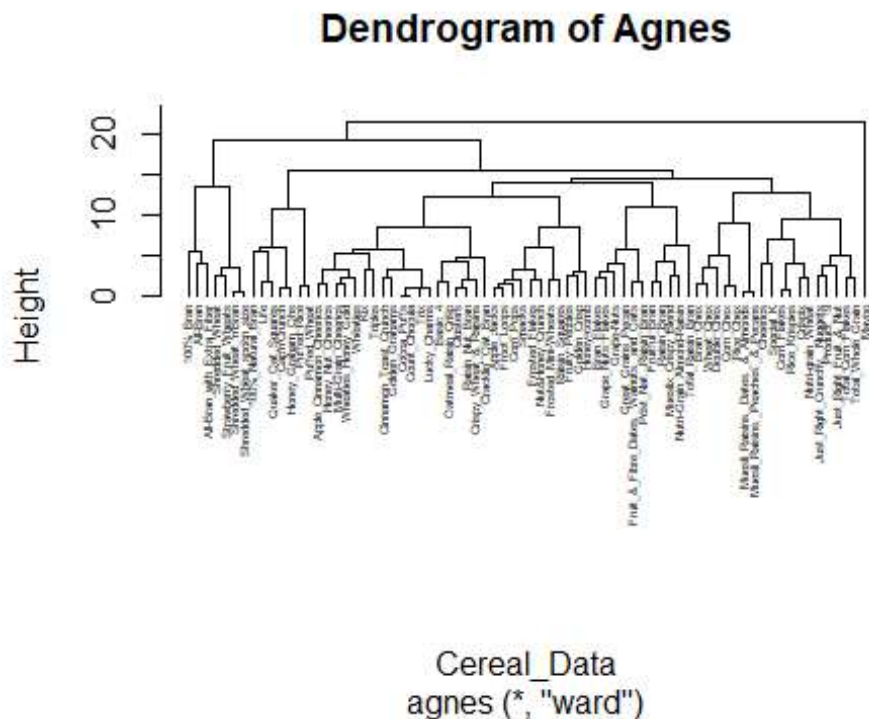
HClust_Ward\$ac

```
## [1] 0.8853988
```

Ward's method produced the highest agglomerative coefficient (.88) this method will be used moving forward

```
## Display the dendrogram produced by Ward's method
```

```
pltree(HClust_Ward, cex = 0.4, hang = -1, main = "Dendrogram of Agnes")
```



Part B

```
## Perform hierarchical clustering using Ward's method and Euclidean distance
```

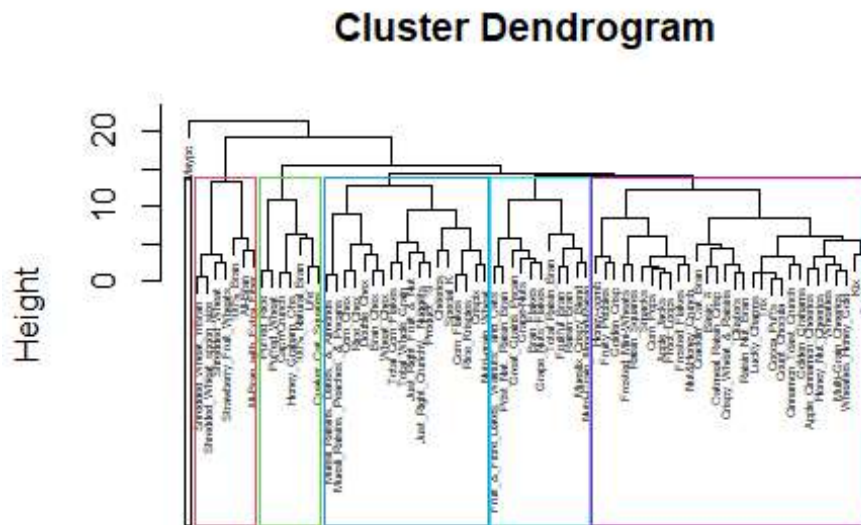
```
Full Cluster <- hclust(Data dist, method = "ward.D2")
```

```
## Plot the dendrogram to determine an appropriate number of clusters
```

```
plot(Full Cluster, cex = 0.4)
```

Divide the dendrogram into six clusters

```
## Six clusters were chosen because they provided meaningful separation
rect.hclust(Full_Cluster, k = 6, border = 1:6)
```



Data_dist
hclust (*, "ward.D2")

```
Cut_Cluster <- cutree(Full_Cluster, k = 6)
```

```
## Display the number of cereals contained in each cluster
table(Cut_Cluster)
```

```
## Cut_Cluster
##  1  2  3  4  5  6
##  7  7 30 18 11  1
```

Six clusters were selected based on the dendrogram produced using Ward's method. The dendrogram showed several large vertical jumps, indicating natural separation between groups. Cutting the tree at six clusters provided a good balance between preserving distinct differences among cereals and avoiding excessive fragmentation into very small clusters. Fewer than six clusters would combine cereals with noticeably different nutritional profiles, while more than six clusters would create clusters that are overly specific and contain only a few observations. Therefore, six clusters provided the most meaningful and interpretable grouping of the cereals.

Part C

```
## Randomly divide the data into two partitions to assess cluster stability
Index <- sample(1:nrow(Cereal_Data), .5*nrow(Cereal_Data))
Part_A <- Cereal_Data[Index,]
```

```

Part_B <- Cereal_Data[-Index,]

## Cluster partition A using Ward's method
A_dist <- dist(Part_A, method = "euclidean")
A_Cluster <- hclust(A_dist, method = "ward.D2")
A_Cluster <- cutree(A_Cluster, k = 6)

## Examine the number of cereals assigned to each cluster
table(A_Cluster)

## A_Cluster
##  1  2  3  4  5  6
## 10  5 14  1  5  2

## Calculate the centroid (mean value) for each cluster in partition A
A_Centroids <- aggregate(Part_A, by = list(Cluster = A_Cluster), mean)

## Remove the cluster labels and convert centroids to a matrix
Centroid_Matrix <- as.matrix(A_Centroids[, -1])

## Calculate distances between each observation in partition B and each cluster centroid
Distance_Matrix <- as.matrix(dist(rbind(Centroid_Matrix, Part_B)))

## Extract only the distances from centroids to observations in B
Distance_B <- Distance_Matrix[1:nrow(Centroid_Matrix), (nrow(Centroid_Matrix) + 1):nrow(Distance_Matrix)]

## Assign each observation in partition B to the nearest centroid
B_Assignment <- apply(Distance_B, 2, which.min)

table(B_Assignment)

## B_Assignment
##  1  2  3  5  6
## 11  5 16  2  3

## Examine the cluster assignments for observations in partition B
table(B_Assignment, Cut_Cluster[-Index])

##
## B_Assignment 1 2 3 4 5
##           1 0 2 4 4 1
##           2 2 0 0 1 2
##           3 0 3 9 4 0
##           5 0 0 0 2 0
##           6 3 0 0 0 0

```

To assess cluster stability, the data were randomly divided into two partitions. Cluster centroids obtained from partition A were used to assign observations in partition B, and

these assignments were compared with the six-cluster solution obtained using the entire dataset. The comparison revealed moderate agreement between the two sets of assignments. While some clusters, particularly Cluster 3, showed reasonable consistency, many observations were assigned to different clusters when the full dataset was used. This suggests that the six-cluster solution captures some meaningful structure in the data, but the clusters are not highly stable and may be sensitive to the particular sample used. Therefore, the clustering solution exhibits moderate rather than strong stability.

Part D

Add the cluster labels to the standardized cereal data

```
Cereal_Data <- cbind(Cereal_Data, Cluster = as.factor(Cut_Cluster))
```

Compute the mean value of each variable within each cluster

```
Cluster_Means <- aggregate(.~ Cluster, data = Cereal_Data, mean)
```

Display the cluster means for interpretation

```
Cluster_Means
```

```
## Cluster  calories    protein      fat    sodium    fiber
## 1         1 -1.5059631  0.5849997 -0.70944304 -1.02099721  1.8138850
## 2         2 -0.6420773 -0.2117484  0.70944304 -0.68443548 -0.4261873
## 3         3  0.1162225 -0.5083157  0.06621468 -0.01648989 -0.4438721
## 4         4  0.2058106  0.3489262 -0.22071561  0.67955045 -0.2329647
## 5         5  0.7453757  0.4522084  0.18058550  0.19658742  0.7903113
## 6         6 -0.3541153  1.3817478  0.00000000 -1.96164410 -0.8977815
##           carbo      sugars      potass  vitamins    shelf    weight
## 1 -0.4811776549 -1.1062791  1.38144582 -0.6624252 -0.2598542 -0.35913268
## 2 -0.8115532407 -0.2214330 -0.41236375 -0.6624252  0.4269033 -1.13201038
## 3 -0.3159898620  0.5563577 -0.42580053 -0.1818422 -0.2598542 -0.07481302
## 4  1.0116305473 -0.6238013 -0.28471438  0.7526247 -0.1263180 -0.09219501
## 5 -0.0006313484  0.4340086  1.01498829  0.1239833  0.8327146  1.32206677
## 6  0.3264071102 -0.9424187 -0.04957081 -0.1818422 -0.2598542 -0.20083243
##           cups      rating      mfrA      mfrG      mfrK      mfrN
## 1 -0.75839139  2.03393380 -0.1162476 -0.6460338 -0.05385697  2.5590737
## 2  0.10221085 -0.04176658 -0.1162476 -0.6460338 -0.66699780 -0.2673659
## 3  0.03978688 -0.56392488 -0.1162476  0.6577799 -0.02319992 -0.2673659
## 4  0.57291583  0.12716434 -0.1162476 -0.2838633  0.28677683 -0.2673659
## 5 -0.69723464 -0.01869160 -0.1162476 -0.4484863  0.11336326 -0.2673659
## 6  0.75675340  0.88922515  8.4860776 -0.6460338 -0.66699780 -0.2673659
##           mfrP      mfrQ      mfrR      typeC      typeH
## 1 -0.36958143 -0.3210386 -0.3210386  0.1162476 -0.1162476
## 2 -0.36958143  3.0727976 -0.3210386  0.1162476 -0.1162476
## 3 -0.06570337 -0.3210386 -0.3210386  0.1162476 -0.1162476
## 4 -0.36958143 -0.3210386  0.9987866  0.1162476 -0.1162476
## 5  1.28793530 -0.3210386 -0.3210386  0.1162476 -0.1162476
## 6 -0.36958143 -0.3210386 -0.3210386 -8.4860776  8.4860776
```



```
## Export cluster means for further examination. File added to GitHub along with assignment  
write.csv(Cluster_Means, "Cluster_Means.csv", row.names = FALSE)
```

After examining the average values of the six clusters, Cluster 1 was identified as the healthiest group of cereals. This cluster exhibited below-average calories, fat, sodium, and sugar levels while maintaining above-average amounts of protein, fiber, and potassium. In addition, cereals in this cluster received the highest consumer ratings among all clusters. These nutritional characteristics make Cluster 1 the most appropriate choice for elementary school cafeterias, as the cereals in this group best support a healthy diet.

Yes, the data should be normalized before performing hierarchical clustering. Variables such as sodium, potassium, calories, and sugar are measured on different scales. Without normalization, variables with larger magnitudes would dominate the Euclidean distance calculations and disproportionately influence the clustering results. Standardizing the variables using Z-scores ensures that each variable contributes equally to the cluster formation process.